

The Utility Problem

Traditionally, electricity consumers paid a flat rate per kWh - consumed energy. With the advancements in household electrification and with changes in society's behaviour, a clear usage pattern is emerging where electricity demand increases during the morning and afternoon hours (often referred to as peak hours) and decreases significantly during mid-day and at night (off-peak hours).

Managing the grid's demand is a complex task with which utilities need to deal on a daily basis. For example, when utilities need to respond to the maximum electricity demand, they have to increase their capacity even though most of this capacity is not required during off-peak hours. Local energy load peaks also contribute to lower power quality which is not optimal for electricity users. The presence of grid peaks will continue to be more prominent with the transition to EVs, electric stoves and electric heating since more tools, equipment and things start to use electricity as a source of energy.

To reduce the need for more capacity, utilities have been introducing different rate plans that aim to shift some of the electricity usage from peak hours to off-peak hours - these plans are called Time-of-Use (ToU) rates. Such rate plans are introduced by more utilities across the globe.

For utilities, to be sure that the ToU rates are impactful, it is important to be able to analyze them in order to predict their impact on the temporal shape of consumption load (peaks and valleys) and on utilities' income. Thus, it is important to have a data platform which allows for such type of analysis.

The Data Solutions and Methods

In general, data needed for such analysis can be summarized in the following list:

1. End consumer consumption data.
2. End consumer information from CIS (Client Information System).
3. GIS data with the end consumer location and connectivity (optional).

The Awesense Energy Transition Platform model all required data for this analysis, and such data can be ingested into the platform for further use.

To conduct such analysis, the analyst would use Awesense Platform APIs (SQL API or REST API) to retrieve the necessary time series and aggregate them to create a current load profile (for the whole grid or sections thereof). The aggregation can be conducted by geographical area using spatial functions, or by grid segment/part (for example, feeder) using `grid_get_connected()` function.

Analysts can analyze the change in consumer behaviour from the introduction of ToU rates (e.g. shifting some consumption to different hours) and simulate the impact of this on the overall

Time Of Use Rate Analysis

[UC49-01] - Awesense Energy Transition Platform Use Case

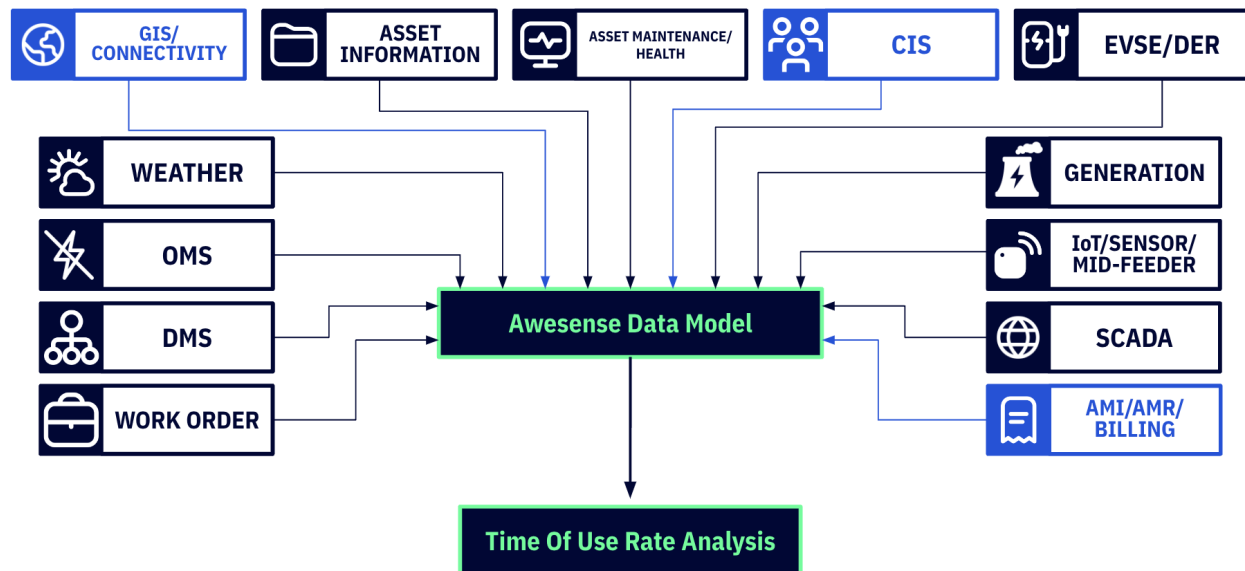


consumption load profile. Multiple scenarios with different assumptions can be developed. Next, the analyst can evaluate the impact of such plans on the profitability of the grid by comparing the income with a flat rate to the income with ToU rate plans. Furthermore, optimization techniques can be applied to optimize consumption such that required capacity is minimized and profit is maximized.

Awesense Platform offers a wide palette of benefits for conducting such analysis, making the analyst job much easier.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter Notebook can be provided upon request.